

Final Project Guidelines

Group Research Project and Grading Rubric

Dr. Abdelrahman Elfikky | Fall 2026 | Teams of 4–5 students

These are the main guidelines for your group project. Your group will study its assigned paper or project, understand the accompanying code, reproduce an agreed-upon result or component, and communicate its findings and contributions professionally.

Each of the seven assignments will be sent separately with detailed instructions, submission requirements, and grading criteria.

Assignment schedule and grade distribution

Assignment	Deadline or schedule	Points	Weight
1 Weekly Progress Reporting and Google Feedback Sheet	Weekly, Sep 25–Dec 11	40	20%
2 Paper Understanding and Reproduction Plan Report	Oct 9 proposed; confirm in assignment	20	10%
3 Mid-Project Meeting and Understanding Evaluation	After about five weeks; 20 minutes per group	20	10%
4 Technical Progress Report, Presentation Draft, and Complete Final Draft	November 27	20	10%
5 Final Presentation and Questions	December 2 or December 7	30	15%
6 Final Delivery Package	December 11	60	30%
7 Presentation Attendance and Peer Review of Three Other Groups	Presentation sessions; reviews due Dec 11	10	5%
Total		200	100%

Percentages apply to the project grade, not the overall course grade. November 27 is the complete-draft deadline and replaces the earlier December 1 deadline.

Scope of reproduction

Groups and papers or projects have already been assigned. For papers longer than 15 pages, an instructor-approved portion may be reproduced. You must still understand the overall paper and supplementary material relevant to that portion.

Immediate priorities and mentoring

Before submitting your first weekly feedback, read your main assigned paper, summarize its main concepts, meet with your group, and select a team leader. Discuss your understanding, identify questions, and agree on initial responsibilities.

Begin with these tasks

- Read your main assigned paper. Identify the research problem, central idea, main concepts, methods or models, and key findings. Note unfamiliar terminology and questions.
- Summarize what you understand in your own words. Explain the paper's purpose and approach, what you have learned so far, and what remains unclear.
- Meet with your team. Discuss your initial understanding, compare questions, select a team leader, and agree on a weekly meeting schedule and initial responsibilities.
- Complete the Google Doc that will be shared with your name. This is your individual reflection and must represent your own understanding, learning, and questions.

Your assigned mentor

Each group will have an assigned mentor. Information about your group's mentor will be shared in the Google Sheet by Monday. Use the designated course channels to communicate your progress, questions, and support needs. Other students and mentors may also provide guidance and support.

Individual reflection and group reporting

The individually named Google Doc is separate from the consolidated group progress update. Complete your own reflection even if some questions overlap with the questions your group answers. You may discuss the paper together, but each student must explain their own understanding in their own words.

In your individual reflection, summarize the main concepts you understood, terminology and models learned, time invested, your contribution to the team discussion, remaining questions, and your next steps. The first weekly submission is due Friday, September 25.

The team leader coordinates the group update; the leader does not write the individual reflections for other members. Individual reflections support the weekly learning and participation assessment within Assignment 1; they are not an additional graded assignment.

Project name and logo

Begin brainstorming a meaningful name for your proposed project and a simple, original logo that represents its purpose. Discuss your ideas with your team and explain how the name and logo connect to the problem you are addressing.

Consider a company owner perspective

For the final delivery package, we may ask you to imagine that your team owns a company developing this project. Consider what problem your product or service would solve, who would use it, how they would benefit, what would distinguish it from existing solutions, and how you would present it to potential customers or partners.

Detailed requirements will be provided in the final-package assignment. For now, prioritize understanding the paper and developing your technical work. The name, logo, and company perspective should support that work, not replace it. The seven-assignment grading distribution remains unchanged.

Learning and teamwork expectations

Read and investigate every week

Read the assigned paper multiple times. Begin with the research question and main contribution, then revisit the methods, equations, experiments, results, and limitations. Read relevant appendices, supplementary experiments, implementation details, repository documentation, and configuration files where available.

Search for supporting information every week. Consult related papers, foundational methods, official documentation, or relevant tutorials. Record the questions you investigated, sources consulted, and how those resources improved your understanding or advanced the project.

- Maintain a terminology log. Explain newly understood terms in your own words and connect them to the project.
- Maintain a model and method log. Explain the purpose, inputs, outputs, assumptions, and limitations of the models or algorithms you study.
- Connect the paper to the GitHub code. Explain relevant files and functions, implementation choices, and experimental outputs. Running the repository alone is insufficient.

Meet and communicate consistently

Choose a team leader. The leader should email the instructor with the group name, members, assigned paper or project, weekly meeting schedule, and shared workspace links. The leader coordinates communication; every member remains responsible for learning and contributing.

Meet at least once each week. Record attendance, decisions, contributions, and next steps. Use the designated course channels to communicate your understanding, questions, blockers, attempted solutions, and support needs promptly.

Dr. Elfikky will explain research-paper structure in a separate activity after the Wednesday lecture. Attend the activity and apply the guidance to your reading and writing.

Brainstorm and attempt an improvement

Developing a successful novel algorithm is not required. Brainstorming original ideas and making a genuine attempt to improve or extend the assigned work are required.

Document your proposed ideas, select at least one feasible idea to explore, and explain why it might help, how it differs from the original work, what you attempted, how you evaluated it, and what you learned. Identify meaningful next steps and future work.

An unsuccessful attempt can earn credit when thoughtfully justified, documented, and analyzed. A careful reproduction, useful dashboard, well-supported analysis, thoughtful survey, or meaningful extension can demonstrate a strong contribution when accompanied by this effort to innovate.

Cite reused ideas, code, figures, and results. Clearly distinguish the original authors' work from your group's contribution. Every member must be able to explain and defend their work.

Weekly progress reporting

Assignment 1 | 40 points | 20% of the project grade

Submit one consolidated Google feedback sheet update each week, beginning Friday, September 25, through December 11, and complete your individual reflection in the Google Doc shared with your name. Clearly identify each member's learning and contributions. Points are assessed across the full reporting period, not awarded as 40 points each week.

Assessment criterion	Points
Evidence of reading the paper and relevant supplementary material	8
Weekly research and search activity, sources consulted, and lessons learned	8
Explanation of terminology, models, and methods understood	8
Individual reflections, hours invested, contributions, and meeting records	8
Progress evidence, challenges, brainstorming, and next steps	8
Total	40

Required fields in each weekly update

Meeting record. Give the date, attendance, discussion points, and decisions.

Individual hours. For each member, report time spent on reading and research, coding and experiments, meetings, and writing or presentation preparation. Do not double-count time.

Paper and supplementary reading. Identify sections, pages, appendices, or resources studied and summarize the main lessons.

Weekly searches. List questions or topics searched, links or citations, and how the resources helped.

Terminology understood. Explain newly learned terms in your own words and their relevance to the project.

Models and methods learned. Explain the models or algorithms studied, including their operation, assumptions, or limitations.

Code and progress evidence. Identify files or functions explored, completed experiments, results, and supporting links.

Brainstorming and trials. Report ideas, refinements, reasons for exploring them, and any tests completed.

Problems and support. Describe what remains unclear, what failed, what you tried, and specific questions.

Next steps. List concrete tasks, responsible members, and intended outcomes.

More reported hours do not automatically earn a higher grade. Assessment emphasizes demonstrated understanding, meaningful effort, and evidence. If no new terms or models were studied, explain what existing knowledge you deepened instead.

Reports and midpoint evaluation

Assignment 2 Paper understanding and reproduction plan

20 points | 10% | October 9 proposed; confirmed in the separate assignment

Explain the problem, motivation, contributions, methodology, data, evaluation measures, and limitations in your own words. Identify the result or component to reproduce, required resources, responsibilities, timeline, risks, and initial improvement ideas.

Assessment criterion	Points
Understanding of the paper and relevant supplementary material	8
Explanation of terminology, models, and code connections	4
Feasible reproduction plan and responsibilities	4
Initial improvement ideas and rationale	4
Total	20

Assignment 3 Midpoint meeting and understanding evaluation

20 points | 10% | After approximately five weeks

Attend a 20-minute TA meeting as a complete group. Every member must explain the paper, relevant code, progress, challenges, and individual contributions. Questions may be directed to any member; the leader must not answer on everyone's behalf.

Assessment criterion	Points
Individual understanding of the paper, terminology, and models	8
Explanation of code and progress evidence	6
Discussion of challenges, proposed ideas, and next steps	6
Total	20

Assignment 4 Technical report and complete drafts

20 points | 10% | November 27

Submit the technical progress report, draft presentation slides, and a complete draft of the dashboard with documentation or the research-style report. Include methods, results, comparisons, limitations, and contributions. This is a complete, reviewable submission, not an outline.

Assessment criterion	Points
Technical progress and results analysis	6
Complete final-product draft	6
Presentation draft	4
Documented brainstorming and progress on an improvement attempt	4
Total	20

Final presentation and delivery

Assignment 5 Final presentation and questions

30 points | 15% | December 2 or December 7

Every member must present and answer questions. Explain the research problem, methods, code, reproduction process, results, limitations, improvement attempt, and future work. Demonstrate the project where appropriate. Duration and group schedules will be announced separately.

Assessment criterion	Points
Technical understanding of the paper, supplementary material, and code	10
Reproduction evidence and interpretation of results	7
Brainstorming, improvement attempt, and lessons learned	5
Presentation organization and delivery	3
Individual participation and responses to questions	5
Total	30

Assignment 6 Final delivery package

60 points | 30% | December 11

Incorporate presentation feedback and submit one of the following final products:

- Professional dashboard: A functional, clearly organized dashboard presenting the project and results, with technical documentation and access or run instructions.
- Research-style report: A structured report explaining the assigned work, reproduction, findings, and contribution. A survey-style report is welcome when it includes original synthesis, critical analysis, citations, and group-proposed future work. It must also satisfy the agreed reproduction scope and innovation-attempt requirement.

Both pathways must include organized code or notebooks, setup instructions and dependencies, reproducibility guidance, results and evidence, references, limitations, future work, each member's contribution, and a brief response to presentation feedback.

Begin considering a project name and logo. A company-owner perspective may also be requested; any such requirements will be explained in the separate final-package assignment.

Assessment criterion	Points
Quality of the professional dashboard or research-style report	20
Reproducibility and documentation	15
Evidence and critical analysis	10
Documented innovation attempt and future work	10
Citations, contribution statement, and response to feedback	5
Total	60

Presentation attendance and peer review

Assignment 7 | 10 points | 5% of the project grade

Each student must attend the designated presentation sessions and individually review three other groups. Attend during the December 2 and December 7 presentation schedule as designated; submit reviews by December 11. Review assignments and the feedback form will be provided separately.

Assessment criterion	Points
Attendance and attentive participation during designated presentation sessions	4
Three substantive peer reviews worth 2 points each	6
Total	10

Required content for each review

- Identify the group name and project title.
- Briefly explain the project and what you learned from the presentation.
- Describe one specific strength, supported by an example.
- Offer one constructive suggestion for improvement.
- Ask one thoughtful technical question or propose a future-work idea.

Reviews must be specific, respectful, and constructive. Your peer-review grade reflects the quality of your feedback, not whether you give another group a positive rating.

Before the final deadlines

- By November 27, submit the technical report, presentation draft, and complete final-product draft.
- Present on your assigned date, December 2 or December 7, and record feedback received.
- By December 11, submit the revised final delivery package and your three individual peer reviews.

Use these guidelines to organize your work now. Each of the seven assignments will be sent separately with its full requirements and submission instructions.